

Microbiology Mid-Term 2023-2024

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MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

- 1) _____ was the first to describe microorganisms, while _____ was the first person to see bacteria.
A) Antoni van Leeuwenhoek / Robert Hook
B) Antoni van Leeuwenhoek / Robert Koch
C) Robert Hooke / Antoni van Leeuwenhoek
D) Robert Koch / Antoni van Leeuwenhoek

- 2) Robert Koch contributed to the field of microbiology by being the first person to
A) develop the tuberculin test only.
B) formulate four postulates for definitively linking a specific microorganism to a specific disease only.
C) use agar as a solidifying agent in growth media only.
D) develop the tuberculin test, formulate four postulates for definitively linking a specific microorganism to a specific disease, and use agar as a solidifying agent in growth media.

- 3) What was Pasteur's purpose in using a swan-necked flask rather than a traditional Erlenmeyer flask for his experiments on spontaneous generation?
A) to prevent air from entering
B) to be able to boil and sterilize the liquid medium
C) to prevent large debris and microbes from entering
D) all of the above

- 4) Which of the following is/are characteristic of all cellular organisms?
A) Communication
B) Evolution
C) Motility
D) Communication, evolution, and motility

- 5) Which individual compared rRNA sequences to generate the tree of life, establishing the three-domain system?
A) Frederick Griffith B) Carl Woese C) Robert Hooke D) Robert Koch

- 6) Microorganisms play key roles in the cycling of important nutrients in plant nutrition, particularly those of
A) carbon only.
B) nitrogen only.
C) sulfur only.
D) carbon, nitrogen, and sulfur.

- 7) Griffith's transformation in *Streptococcus pneumoniae* led to the discovery of
A) DNA as the molecular basis of heredity.
B) mRNA as the template for protein synthesis.
C) polysaccharides as the major component of capsules.
D) Archaea as the third domain of life.

- 8) An organism of the genus *Staphylococcus* is_____, while an organism of the genus *Escherichia* is_____.
- A) spherical / rod shaped
 B) rod shaped / coiled
 C) spherical / coiled
 D) soiled / spherical
- 9) When medical devices are left in the body, bacteria may grow on them as_____, which makes them especially resistant to treatment.
- A) biofilms
 B) liquids
 C) populations
 D) communities
- 10) *Archaea* and *Bacteria* are unified as prokaryotes in lacking ____ which *Eukarya* contain, such as mitochondria.
- A) membranes
 B) nuclei
 C) membrane-enclosed organelles
 D) nuclei and membrane-enclosed organelles
- 11)The cell wall of a gram-positive bacterium is composed of a thick ____ layer.
- A) protein
 B) poly- β -hydroxybutyric acid (PHB)
 C) lipopolysaccharides (LPS)
 D) peptidoglycan
- 12) Which of the following statements is TRUE?
- A) Fimbridd are generally longer and less numerous than flagella.
 B) Fimbriae are involved in genetic exchange between cells.
 C) Hami are common in Bacteria but not found in Archaea.
 D) Fimbriae are usually shorter than flagella, whereas pili are involved in genetic exchange.
- 13) Transport proteins located in the cytoplasmic membrane are necessary when ____
- A) diffusion will not allow adequate amounts of a substance to enter the cell.
 B) movement into the cell is against a concentration gradient.
 C) the level of nutrients in nature is very low.
 D) nutrient concentration is very low in the environment, transport is needed against the concentration gradient, or diffusion is not possible.
- 14) Based on the table of attributes given below, which of the following statements are FALSE about the two organisms?

Characteristic	Bacterium A	Bacterium B
Endospore formation	yes	no
Capsule	no	yes
Type IV pili	yes	no
Flagella	no	no
Morphology	bacillus	bacillus

- A) Bacterium A is more resistant to heat and ultraviolet light.
- B) Bacterium B likely forms a slime layer better than Bacterium A.
- C) Bacterium B is likely to exhibit motility.
- D) Both bacteria may attach to surfaces.

15) A chemoorganotroph and a chemolithotroph in the same environment would NOT compete for ____

- A) oxygen.
- B) carbon.
- C) nitrogen.
- D) phosphorous.

16) If ΔG° is positive, the reaction is ____

- A) exergonic and requires the input of energy.
- B) endergonic and requires the input of energy.
- C) exergonic and energy will be released.
- D) endergonic and energy will be released.

17) If an oxidation reaction occurs, ____

- A) simultaneous reduction of a different compound will also occur, because electrons do not generally exist alone in solution.
- B) another oxidation reaction will occur for a complete reaction, because one oxidation event is considered a half reaction.
- C) a cell is undergoing aerobic respiration, because oxygen is being used.
- D) a reduction reaction would not occur, because they are opposite reaction mechanisms.

18) An organism aerobically respire glucose to obtain energy. Glucose is the ____ and oxygen is the ____

- A) Electron donor/electron acceptor.
- B) Electron donor/electron donor.
- C) Electron acceptor/electron donor.
- D) Electron acceptor/electron acceptor.

19) An organism that oxidizes hydrogen and fixes CO₂ is a ____

- A) Chemoorganoheterotroph.
- B) Photoorganoheterotroph.
- C) Chemolithoautotroph.
- D) Photolithoautotroph.

20) Enzymes catalyze chemical reactions by ____

- A) lowering the activation energy, thus increasing the rate of the reaction.
- B) increasing the activation energy, thus increasing the rate of the reaction.
- C) making the reaction exergonic.
- D) releasing free energy.

21) Which statement is FALSE regarding enzymes?

- A) All enzymatic reactions are theoretically reversible.
- B) Endergonic reactions are coupled with exergonic reactions to gain enough energy to occur.
- C) Prosthetic groups and coenzymes are small nonprotein molecules that are a part of many enzymes.
- D) The part of the enzyme where the substrate binds is called the redox site.

22) The exact composition of _____ media is known. _____ media inhibit the growth of some microbes. _____ media are used to culture fastidious microbes, particularly pathogens.

- A) complex / Differential / Selective
- B) complex / Enriched / Differential
- C) defined / Selective / Enriched
- D) defined / Enriched / Differential

23) The time interval required for the formation of two cells from one is called the _____

- A) generation time.
- B) growth time.
- C) growth rate.
- D) division rate.

24) When attempting to determine viable cell counts of a heat-sensitive bacterial isolate, which cell counting method should be avoided?

- A) Spread-plate method
- B) Pour-plate method
- C) Live staining and direct counting with a microscope
- D) Both spread- and pour-plate methods

25) A microbe growing in a refrigerator is likely _____

- A) psychophilic.
- B) mesophilic.
- C) psychrotolerant or psychrophilic.
- D) hyperthermophilic.

26) If heterotrophic bacteria were grown in a minimal glucose medium and transferred during exponential growth phase into complex LB medium, the cells would have a relatively _____

- A) short exponential phase because it already is in exponential growth phase.
- B) high growth yield compared to cells transferred during stationary growth phase.
- C) long lag phase because it has to adjust to a different chemical environment/medium.
- D) short lag phase because of the abundance of available nutrients.

27) Which of the following statements is FALSE?

- A) In the death phase, bacterial growth may cease as a result of oxygen and nutrient depletion.
- B) The generation time of bacteria may vary from species to species.
- C) Bacteria typically divide by binary fission, producing two daughter cells.
- D) In the lag phase, cell death exceeds cell division.

28) A bacterium possessing _____ will produce O₂ bubbles when placed in the presence of hydrogen peroxide.

- A) catalase
- B) superoxide dismutase
- C) superoxide reductase
- D) peroxidase

29) The sigma subunit of RNA polymerase functions to

- A) form RNA polymerase-DNA complexes.
- B) locate, recognize, and bind to promoter regions.

- C) terminate mRNA synthesis of genes at their stop sites.
- D) do both A and B.

30) Which statement below provides an explanation of why a lagging strand (rather than two leading strands) exists during DNA replication?

- A) New nucleotides can be added only to a 3' (-OH) end.
- B) Only one primer can exist at a time.
- C) There is only one DNA polymerase per cell.
- D) There is only one replisome per cell.

31) DNA-binding proteins interact predominantly within which portion of a double-stranded DNA helix?

- A) Major groove
- B) Minor groove
- C) 3' end
- D) Supercoil

32) Supercoiling is important for DNA structure because _____

- A) it holds together the antiparallel strands of DNA in the double helix.
- B) it provides energy for transcription.
- C) it condenses the DNA so that it can fit inside the cell.
- D) it prevents RNA from pairing with DNA in the double helix.

33) Many pharmaceutical drugs specifically inhibit transcription in *Bacteria* but not *Archaea* or *Eukarya*. Why would drugs that inhibit transcription in only affect *Bacteria* and not *Archaea* even though they are both prokaryotes?

- A) Archaea and Eukarya have very similar ribosomes that are different than bacterial ribosomes.
- B) Bacteria lack a nucleus.
- C) Archaea lack operons.
- D) Archaea and Eukarya have very similar RNA polymerases that are different than bacterial RNA polymerases.

34) How are plasmids different than chromosomes?

- A) Plasmids are always small, linear pieces of DNA.
- B) Plasmids are composed of single-stranded DNA.
- C) Plasmids contain genes that are NOT essential for cellular growth and replication.
- D) Plasmids carry unimportant genes that are of little significance for the ecology and metabolism of an organism.

35) Transfer RNA (tRNA) molecules _____

- A) function to transfer ribonucleotides to RNA polymerase during transcription.
- B) function to transfer the correct amino acids to the ribosome during translation.
- C) contain codons that bind to ribosomes during translation.
- D) are only present in the nucleus or eukaryotes.

36) You are studying a protein in *Salmonella typhimurium* that you believe is a toxin. Whenever you attempt to purify the protein from lysed cell cultures, you get two forms of the protein. One form is smaller than the other and is missing 15 amino acids from the N-terminus compared to the larger form. This leads you to hypothesize that _____

- A) there are two termination sites in the mRNA.
- B) the protein requires chaperonins to fold properly.

- C) the protein is secreted and folds outside of the cell.
- D) you need to redo the experiment because there should only be one form.

37) Viral genomes can be _____

- A) linear or circular.
- B) double-stranded or single-stranded.
- C) composed of RNA or DNA.
- D) all of the above.

38) Viral replication is _____

- A) independent of the host cell's DNA but dependent on the host cell's enzymes and metabolism.
- B) independent of both the host cell's DNA and the host cell's enzymes and metabolism.
- C) dependent on the host cell's DNA and RNA.
- D) dependent on the host cell's DNA, RNA, enzymes, and metabolism.

39) Viral replication occurs _____

- A) intracellularly.
- B) extracellularly.
- C) both intracellularly and extracellularly.
- D) either intracellularly or extracellularly, depending on the virus involved.

40) Cellular receptors may be composed of _____

- A) proteins.
- B) carbohydrates.
- C) lipids.
- D) combinations of proteins, carbohydrates, and/or lipids.

41) A prophage replicates _____

- A) along with its host while the lytic genes are expressed.
- B) along with its host while the lytic genes are not expressed.
- C) independently of its host while the lytic genes are expressed.
- D) independently of its host while the lytic genes are not expressed.

42) When a solution composed of bacteria and infectious virions are mixed and spread on an agar plate, _____ form where viruses lyse the host cells.

- A) insertion sequences
- B) plaques
- C) prophages
- D) colonies

43) The growth of viruses in a culture is described as a one-step growth curve, because _____

- A) virion numbers show no increase during intracellular replication and can only be counted after the virions burst from the host cell.
- B) there is only one step in the viral life cycle which leads to only one replicative cycle in a culture.
- C) assembly and release actually occur in one step.
- D) the eclipse phase prevents the plating and enumeration of virions although new virions are produced at a steady rate during the eclipse phase.

44) Attenuation is a transcriptional control mechanism that relies on _____

- A) attenuator molecules binding to mRNA.
- B) effector proteins to control the binding of RNA polymerase.
- C) transcriptional feedback inhibition.
- D) the secondary structure of the mRNA leader.

45) The _____ is a bacterial regulatory mechanism used to survive nutrient-poor conditions, antibiotic exposure, and environmental stresses.

- A) catabolite repression
- B) heat shock response
- C) stringent response
- D) two-component system

46) When histidine is added to a culture already growing exponentially in a medium without histidine, what occurs?

- A) All cellular growth ceases.
- B) Growth continues, but the production of enzymes required for the synthesis of histidine ceases.
- C) Growth continues, but the production of enzymes required for the synthesis of histidine increases.
- D) The cell returns to the lag stage of growth to synthesize the proteins necessary for the metabolism of histidine.

47) In negative control of transcription by the lac operon, how does the presence of an inducer affect transcription?

- A) The inducer binds to the operator.
- B) The inducer does not bind to the operator.
- C) The inducer causes the repressor to bind to the operator.
- D) The inducer prevents the repressor from binding to the operator.

48) What occurs when an inducer is added to an environment containing an organism with a metabolic pathway controlled by a repressor?

- A) The inducer combines with the repressor and activates the pathway.
- B) The inducer combines with the repressor and inactivates the pathway.
- C) The inducer combines with the substrate and blocks induction.
- D) The inducer does not combine with, but functions as a chaperone molecule for, the enzyme-substrate complex.

49) Which of the following do NOT bind to the promoter sequence during regulation?

- A) Activators
- B) Inducers
- C) Repressors
- D) None of these binds directly to the promoter sequence.

50) The most frequent way in which small regulatory RNA molecules exert their effects is by _____

- A) base pairing with other RNA molecules that have regions of complementary sequence.
- B) binding to a repressor and repressing enzyme transcription.
- C) acting as an inducer that then binds to an activator protein to allow transcription to proceed.
- D) signal transduction.

51) Pathogenic bacteria such as *E. coli* 0157:H7 and *Staphylococcus aureus* have a strong selection pressure to refrain from activating an immune response until their numbers are great. _____ is thus an effective way these

pathogens control production of certain virulence factors.

- A) Attenuation
- B) Differentiation
- C) Signal transduction
- D) Quorum sensing

参考答案

1-5	CDCBB	6-10	DAAAD	11-15	DDDCB	16-20	BAACA	21-25	DCABC
26-30	DDADA	31-35	ACDCB	36-40	CDAAD	41-45	BBADC	46-51	BDADAD